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Project Title:	NSCOR: The contribution of non-targeted effects in HZE cancer risk		
Division Name:	Human Research		
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Comments:	NOTE: Former affiliation was Lawrence Berkeley National Laboratory; moved to NYU in August 2008. Moved to UCSF September, 2015		
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COI Name (Institution):	Costes, Sylvain (Lawrence Berkeley National Laboratory) Demaria, Sandra (NYU Medical Center) Mao, Jian-Hua (Lawrence Berkeley National Laboratory) Balmain, Allan (University of California San Francisco)		
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Task Description:	Clear mechanistic understanding of the biological processes elicited by high linear energy transfer (LET) radiation increased cancer risk will reduce the uncertainties in assessing the risk of space flight. Classic targeted radiation effects such as cell kill and mutation are proportional to dose and their LET dependence is described by relative biological effectiveness (RBE). However recent data suggests that high LET radiation affects multicellular processes that promote cancer. We propose that radiation carcinogenesis results from two classes of ionizing radiation effect: targeted effects that alter genomic sequence and non-targeted effects (NTE) that alter phenotype and multicellular interactions. Since NTE are not proportional to dose, new approaches are required to assess the impact of NTE, which alter gene expression, cell signaling, genomic stability, and cell phenotypes, on carcinogenesis. It is unknown to what extent NTE contribute to the significantly greater carcinogenic effect of high LET radiation. We hypothesize that NTE promotes epithelial cancer. Project 1 will test use novel transgenic and humanized mouse mammary models to evaluate the relative contribution of radiation targeted and NTE to carcinogenesis as a function of radiation quality and of age a radiation chimera model. We will monitor inflammation as a key process affected by radiation NTE and conduct a systems biology analysis of mammary gland. Project 2 will employ new systems genetic analysis of cancer susceptibility in radiation sensitive organs to define critical regulatory networks that impact the relative carcinogenic effectiveness of high LET radiation. Project 3 will use Monte Carlo simulations of targeted radiation effects and agent based modeling of cell interactions to model the relative contributions of targeted and NTE to carcinogenesis. This NSCOR (NASA Specialized Center of Research) will provide important data on radiation quality dependence of targeted and non-targeted radiation effects and the genetic determinants of cancer susceptibility, which together will be used to reduce uncertainties in NASA radiation risk assessment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	First, radiation can be considered an experimental probe that perturbs tissue function. Studying these perturbations is a means to define critical biological response to stress. Second, we have develop a comprehensive database of normal tissue and irradiated tissue by analyzing different organs in each mouse. This database will become a public resource. Third, our studies will identify genetic predisposition to radiation associated cancer, and may thus help protect humans who have the greatest risk of cancer. Fourth, our studies have provided compelling evidence of radiation action via non-targeted effects, which may inform how radiation risks are estimated and managed prospectively.
	This project used experimental studies and modeling to define the efficiency and physiological context in which high LET (linear energy transfer) radiation exposure increases epithelial cancer risk. We hypothesized that targeted and non-targeted effects (NTE) both contribute to radiation carcinogenesis. Targeted effects includes DNA damage responses and changes in genomic sequence that is the result of interaction between ionizing irradiation and DNA. NTE alter the phenotype and multicellular interactions that can contribute to cancer. Our hypothesis proposed that the greater carcinogenic risk from high LET irradiation is due to NTE promotion of targeted radiation effects. A perspective/review was published by the group in 2014. These studies provided a comprehensive analysis of solid tumor frequency in different tissues, analyzed with computational modeling and systems biology. The NSCOR was composed of 3 projects at New York University (NYU), University of California San Francisco (UCSF,), and Lawrence Berkeley National Laboratory (LBNL). Due to a 6 months hiatus as a consequence of Hurricane Sandy at NYU, a no-cost extension was awarded to complete the analysis and manuscript preparation of remaining data. The following is a summary of these studies. The goal of Project 1 was to determine the contribution of NTE as a function of radiation quality on breast cancer. We used a radiation chimera mammary model to discern the effects of radiation-induced microenvironment from DNA damage and potential mutation. In this model we first surgically cleared the mammary gland of endogenous epithelium. Then the mouse was irradiated and the fat-pad subsequently orthotopically transplanted with un-irradiated, non-malignant epithelial cells primed for malignant transformation by Trp53 deletion that were evaluated for tumors for 600 days. A paper published in Cancer Research in 2014 reported results of studies in which mice were irradiated with either 100cGy of gamma-radiation or 11, 30, or 81 cGy from 350MeV/amu Si particles compared to sham-irradiated controls prior to transplantation with Trp53 null mammary fragments. Trp53 null tumors arising in mice densely irradiated had a shorter median time to appearance and grew faster once detected compared to those in sham-irradiated or gamma-irradiated mice. Tumors were further classified by markers keratin 8/18 (K18, KRT18), keratin 14 (K14, KRT18), and estrogen receptor (ER, ESR1) and expression profiling. Most tumors arising in sham-irradiated hosts contained both K18 and K14 positive cells (K14/18) while most tumor arising in irradiated hosts were K18 alone. K14/18 tumors were predominantly ER positive while K18 tumors were predominantly ER negative. Although K18 tumors tended to grow faster and be more metastatic than K14/18 tumors, K18 tumors in Si particle-irradiated mice grew significantly larger compared to controls and were more metastatic compared to sham-irradiated mice. The expression profile that distinguished K18 tumors arising in particle-irradiated compared sham-irradiated mice was enriched in mammary stem cell, stroma, and Notch signaling genes. These data suggest that densely ionizing radiation carcinogenic effects mediated by the microenvironment elicit more aggressive tumors compared to tumors arising in sham-irradiated hosts. To compare the contribution of targeted and NTE, we irradiated mice after Trp53 null outgrowth was established (i.e., epithelium was exposed, i.e., targeted and NTE) and the other group was irradiated before transplantation (i.e., NTE only, epithelium was not exposed). Tumors in mice irradiated with densely ionizing 600 MeV/amu Fe charged particles or sparsely ionizing gamma-ray radiation were compared to those in contemporaneous sham-irradiated controls. Irradiation of the in situ mammary outgrowth with either radiation quality decreased Trp53 null tumor latency and increased tumor frequency a year after. Mammary tumor latency arising from non-irradiated Trp53 null outgrowths in irradiated hosts was also decreased indicating that this effect was mediated by the microenvironment. Tumors arising from densely ionizing irradiated outgrowths gave rise to aggressive tumors with increased growth rate compared to sham-irradiated mice. Importantly, increased tumor aggressiveness was also observed in tumors arising from irradiated hosts, suggesting that this effect was mediated by NTE. Only densely ionizing radiation affected tumor markers of differentiation in both models. The data from these studies supports the conclusion that tumor aggressiveness is indirectly promoted by NTE and that NTE elicited by HZE particles results in more aggressive tumors. The RNA of

these tumors has been sequenced. Notably, preliminary analysis using unsupervised clustering separated tumors arising from irradiated epithelium in irradiated hosts (i.e., targeted) from those that arose in irradiated mice. These data are in preparation for publication.

We also evaluated the effect of age in the radiation chimera-experiment; mice were irradiated at 10 months, simulating the average age for astronauts. Notably high LET exposure of aged mice is much more efficient inducing tumorigenesis than low LET. Cancer incidence in humans increases exponentially with age, with 75% of newly diagnosed cases occurring in susceptible populations aged 55 years or older. The aging process has been shown to be associated with increased levels of chronic inflammation, which are thought to contribute to many age-associated diseases, including cancer, and increased serum levels of IL-6 have been reported in older individuals. Interestingly, A-bomb survivors have significant increases in serum IL-6 levels that are still detectable five decades after exposure. These findings suggested to us the possibility that radiation carcinogenesis, at least in part, is mediated by induction of a pro-inflammatory environment similar to the aging process. If so, we predicted that radiation NTE would be epistatic with age.

To test this, mice were irradiated at 10 months of age, comparable to the middle age of astronauts during space-flight missions to test the hypothesis that NTE overlap with the biology that increases cancer incidence in aged versus young people. The inguinal mammary glands of 200 mice were cleared at 3 weeks of age and the mice housed under standard conditions. Hosts were irradiated at 10 months of age with 10, 50, or 100 cGy gamma-radiation or 350 MeV/amu Ar or Si, or 600 MeV/amu Fe at a fluence of 1 particle per 102 mm, and transplanted 3 days later with Trp53 null epithelium. Tumorigenesis was monitored for 600 days. As expected due to decreased ovarian hormones in aged mice, fewer mammary outgrowths were obtained in aged mice; outgrowth efficiency is >90% in young mice mammary but ~50% in aged mice. This limited the ability to compare dose or particles individually. However, we were able to compare tumors arising in sham-irradiated mice to gamma-irradiated hosts or HZE-irradiated hosts. Age and radiation NTE were not epistatic as both low- and high-LET irradiation of aged hosts significantly increased tumor incidence and tumor growth rate. Moreover, this experiment revealed a strong radiation quality effect in which HZE NTE was considerably more effective than gamma-radiation.

Task Progress:

Expression profiling of these tumors indicate that the top ranked ($p>0.00001$) processes that distinguish tumors from aged mice irradiated with HZE particles from age-matched, sham-irradiated tumors were inflammation (70 genes), immune cell trafficking (79 genes), and cancer (271 genes). Among the 20 most significant upstream regulators ranked by z-score ($p<0.05$), the majority are pro-inflammatory factors such as IL-1ß, TNFalpha, and IL6. Consistent with our previous findings, TGFß1 is predicted by IPA to also be a significant regulator in cancers from both HZE and gamma-irradiated host. We speculate that gamma- and HZE-irradiated microenvironments might evoke common mediators, but their magnitude or persistence might be different. Among the top-ranked upregulated molecules are COX2 and granzyme H. We validated this signature using relative COX2 transcript abundance and protein expression, which support high levels of COX-2 in tumors arising in HZE-irradiated mice. These data are being prepared for publication (Ouyang et al.).

Together these data lead us to postulate that a high degree of immunosuppressive inflammation promotes these aggressive tumors. To test this idea, we used immunostaining to evaluate lymphoid and myeloid cellular subsets within tumors. Most strikingly, tumors arising in HZE-irradiated aged hosts have significantly ($p<0.05$) more myeloid derived suppressor cells (MDSC) and Treg cells compared to tumors arising in age-matched, sham-irradiated hosts. Treg frequency inversely correlates with tumor growth rate and MDSC is positively correlated with Cox2 staining, leading to the new hypothesis that HZE radiation exposure in mid-life is strongly immunosuppressive.

Project 2, directed by Dr. Allan Balmain at UCSF, had the ultimate goal of identifying the genetics associated with susceptibility to cancer risk from high LET radiation. The strategies employed were based on systems genetics approaches to identify germ line polymorphisms and somatic genetic changes associated with tumor development. This project focuses heavily on genetic analysis of radiation-induced tumors. A paper recently published in Molecular Carcinogenesis describes the phenotype of germline deletion of a small N-terminal proline-rich region of Trp53 in an in vivo mouse model (Trp53 deltaP), which causes no spontaneous tumors; a manuscript on this model was published in 2015 by the group. Upon exposure of the Trp53 deltaP mouse on the original mixed background of 129/Sv/C57BL/6 to either low and high LET radiation, a wide range of solid tissue tumors develops, including those of the liver, lung, and kidney. This is in stark contrast to the tumor spectrum in the Trp53 null (-/-) mouse irradiated with 4 Gy low LET, which generates mainly lymphomas and sarcomas. The Trp53 deltaP mouse therefore represents a sensitive model to study radiation-induced tumors similar to those humans.

Interestingly the irradiated tumor spectrum of both Trp53 deltaP and Trp53+/- mice in the FVB/N background falls into three main tumor types: thymic lymphomas, mammary tumors, and skin carcinomas. It should be noted that lymphomas are often observed post-radiation in mice of compromised Trp53 function, particularly after exposure to high radiation doses (4 Gy). After exposure to a low dose (50cGy) of either low or high LET radiation, we observed increased frequency of mammary tumors and skin carcinomas in both genotypes. Although the tumor spectrum is similar between mice exposed to high or low LET radiation, there was an increased incidence of mammary tumors following high LET radiation.

Of note the tumor spectrum in 129/Sv Trp53+/- mice was markedly different from FVB/N. Regardless of radiation quality there was a high incidence of thymic lymphomas. A wide range of tumors of varying types with no distinguishable pattern but no skin carcinomas were observed. A small number of mammary tumors developed following high LET radiation (n=3), whilst none were observed in the gamma-irradiated group. These differences further illustrate the influence of background strain upon tumor latency, susceptibility, and spectrum following exposure to high or low LET radiation. These data demonstrate that FVB/N mice are particularly susceptible to radiation-induced mammary tumor development. High LET radiation was also more potent at inducing a higher incidence of mammary tumors with a shorter latency compared to gamma-radiation ($p=0.0004$ CoxPH).

Little is known about the mechanisms by which high LET induces carcinogenesis. Exposing mice of abrogated Trp53 function to either high or low LET radiation provided us with a unique opportunity to compare and contrast the genetic aberrations and potential radiation-quality specific signatures in tumors. Next generation sequencing offers a powerful tool to gain a deeper understanding of the complex mutational processes driving radiation-induced tumorigenesis. In collaboration with Dr. David Adams at Sanger Institute, Cambridge, exome sequencing was performed on 60 tumor samples, along with 10 control samples. We have sequenced the most frequently observed tumor types, (thymic lymphomas, mammary tumors, and skin carcinomas), in Trp53 deltaP and Trp53+/- on FVB/N or 129/Sv/C57BL/6

	backgrounds induced by either Fe-ion or gamma radiation. Project 3, led by Dr. S. V. Costes and Dr. Mao, undertook to use computational modeling of experimental data. A manuscript was published in Radiation Research that predicts RBE for cell death based on experimental measurements done with X-ray. Briefly, Linear-quadratic models have been used for decades to interpret cell survival after exposure to ionizing radiation. However, parameters for these models change with LET, forcing investigators to measure experimentally cell survival for any LET of interest. In contrast to the current paradigm, we hypothesize that double strand breaks (DSBs) are moved into repair domains and consequently merge into clusters. Overall, this work suggests that microdosimetric properties of ion tracks at the sub-micron level are sufficient to explain both RIF data and survival curves for any LET, similarly to the Local Effect Model assumption. The use of agent-based modeling (ABM) to estimate the contribution of stem cells to carcinogenesis published in 2013 in collaboration with Project 1, was extended to allow integration of the multi-stage clonal expansion model with NTE agent-based approaches. The goal was to simulate tumor data in silico by assuming that tumors arise via successive DNA mutation and that promotion is influenced by non-targeted effects based on the experimental data in Project 1. One challenge in generating tumor in silico is the duration of simulations using our current ABM, therefore we adopted a simpler computer model (automata) that allows cells to be tracked faster over a 60 year simulation (from age 20 to 80 years old) to implement the multi-stage clonal expansion model to simulate tumor incidence arising spontaneously in human population due to random mutations using these automata. Each simulated person is a monolayer of 4 million cells (referred as “tissue”), where cells divide only when neighboring cells have died. Parameter sweeps were done so that simulations predicted cancer incidence matching a normal human population.
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